

Notes on Wronskian

For two meromorphic functions, $u_1(z)$ and $u_2(z)$, defined in a domain S of the complex plane, their Wronskian is defined as

$$W[u_1(z), u_2(z)] \equiv \begin{vmatrix} u_1(z) & u_2(z) \\ u_1'(z) & u_2'(z) \end{vmatrix} = u_1(z)u_2'(z) - u_1'(z)u_2(z).$$

1 Linear dependence

Two functions, $u_1(z)$ and $u_2(z)$, are linearly dependent, if there exist constants c_1 and c_2 , not zero at the same time, such that

$$c_1 u_1(z) + c_2 u_2(z) = 0$$

can be satisfied. If no such constants can be found, the functions are linearly independent.

For linear dependent functions, assume $u_1(z)$ is not identically zero, and define

$$f(z) = \frac{u_2(z)}{u_1(z)}.$$

Then, $u_1(z)$ and $u_2(z)$ are linearly dependent if and only if $f(z)$ is constant. Since

$$f'(z) = \frac{u_1(z)u_2'(z) - u_1'(z)u_2(z)}{u_1(z)^2} = \frac{W(z)}{u_1(z)^2}, \quad (1)$$

we can see that $f(z)$ is constant if and only if $W(z) = 0$. Also, $W(z) \neq 0$ implies linear independence.

2 Wronskian for second order ordinary differential equation

If $u_1(z)$ and $u_2(z)$ are the solutions to the second order ordinary differential equation (ODE)

$$\frac{d^2 u(z)}{dz^2} + p(z) \frac{du(z)}{dz} + q(z)u(z) = 0, \quad (2)$$

then their Wronskian will only depend on the variable z and the function $p(z)$. In fact, take the derivative of the Wronskian with respect to z , we will have

$$\begin{aligned} \frac{dW(z)}{dz} &= u_1(z)u_2''(z) - u_1''(z)u_2(z) \\ &= -u_1(z) [p(z)u_2'(z) + q(z)u_2(z)] + [p(z)u_1'(z) + q(z)u_1(z)] u_2(z) \\ &= p(z) (u_1'(z)u_2(z) - u_1(z)u_2'(z)) = -p(z)W(z). \end{aligned}$$

The Wronskian therefore can be integrated,

$$W(z) = W(a) \exp \left(- \int_a^z p(z') dz' \right).$$

Consider the Bessel equation,

$$\frac{d^2 u(z)}{dz^2} + \frac{1}{z} \frac{du(z)}{dz} + \left(1 - \frac{\nu^2}{z^2} \right) u(z) = 0,$$

which has two linearly independent solutions, $J_\nu(z)$ and $N_\nu(z)$. For $|z| \ll 1$ and $\nu \neq 0$, we have

$$J_\nu(z) \rightarrow \frac{1}{\Gamma(\nu+1)} \left(\frac{x}{2} \right)^\nu, \quad N_\nu(z) \rightarrow -\frac{\Gamma(\nu)}{\pi} \left(\frac{2}{x} \right)^\nu.$$

Then,

$$J'_\nu(z) \rightarrow \frac{1}{\Gamma(\nu)} \frac{x^{\nu-1}}{2^\nu}, \quad N'_\nu(z) \rightarrow \frac{\Gamma(\nu+1)}{\pi} \frac{2^\nu}{x^{\nu+1}},$$

and the Wronskian for $J_\nu(z)$ and $N_\nu(z)$, $\nu \neq 0$, at $|z| \ll 1$, is

$$W(z) = J_\nu(z)N'_\nu(z) - N_\nu(z)J'_\nu(z) = \frac{2}{\pi z}.$$

Therefore, for arbitrary z , we have, starting from $|z'| \ll 1$,

$$W(z) = W(z') \exp \left(- \int_{z'}^z \frac{dz''}{z''} \right) = \frac{2}{\pi z'} \frac{z'}{z} = \frac{2}{\pi z},$$

which does not require z to be around the origin. The similar calculation can be performed for $\nu = 0$.

3 Finding linearly independent solution

Suppose we know a solution $u_1(z)$ to the second order ODE (2), then we can construct a second linearly independent solution $u_2(z)$ from $u_1(z)$. From Eq. (1), we have

$$\frac{d}{dz} \left(\frac{u_2(z)}{u_1(z)} \right) = \frac{W(a)}{u_1(z)^2} \exp \left(- \int_a^z p(z'') dz'' \right).$$

Integrating from a to z , we will have

$$\frac{u_2(z)}{u_1(z)} = W(a) \int_a^z \frac{1}{u_1(z')^2} \exp \left(- \int_a^{z'} p(z'') dz'' \right) dz' + \frac{u_2(a)}{u_1(a)}.$$

Since $W(a) \neq 0$, we obtain the second linearly independent solution as

$$u_2(z) = u_1(z) \int \frac{1}{u_1(z')^2} \exp \left(- \int^{z'} p(z'') dz'' \right) dz'$$

In particular, if $p(z) = 0$, we will have a simple form of the second equation,

$$u_2(z) = u_1(z) \int \frac{dz'}{u_1(z')^2}.$$

For example, the differential equation

$$u''(z) - k^2 u(z) = 0$$

has a solution $u_1(z) = e^{kz}$. The second solution is then, for $k \neq 0$,

$$u_2(z) = u_1(z) \int^z \frac{dz'}{u_1(z')^2} = e^{kz} \int^z e^{-2kz'} dz' = -\frac{1}{2k} e^{kz} e^{-2kz} = -\frac{1}{2k} e^{-kz},$$

which is clearly a second solution to the differential equation and is linearly independent from $u_1(z)$.